

1. **Explain the advantage and disadvantage of road transport. Explain the factors that affects the selection of highway alignment.**

Answer:

Road Transportation is defined as the movement of people and goods from one place to another via roads. It is the most common and vital mode of transportation in Nepal. While it offers significant benefits for connectivity and economic growth, it also faces severe technical and environmental challenges, particularly in hilly and mountainous regions.

Advantages of Road Transport

- i. **High Accessibility and Flexibility:** Road transport provides the highest level of accessibility to almost all potential destinations, including remote hills and mountainous regions. It allows for door-to-door service and flexible travel routes.
- ii. **Economic Efficiency:** The capital cost for road infrastructure is relatively cheaper. It is considered the most economical means of transport overall. Furthermore, proper road network planning minimizes travel distances and vehicle operating costs.
- iii. **Suitability for Short Distances:** Roadways are the most appropriate and efficient means for short-distance travel.
- iv. **Socio-Economic Development:** Roads play a critical role in reducing regional disparities by improving access to essential services such as education, healthcare, and markets.
- v. **Adaptability to Terrain:** In complex topographical areas like Nepal, roadways are often the only feasible means of providing essential connectivity for scattered settlements.

Disadvantages and Challenges of Road Transport

- i. **Environmental Degradation:** Especially in hilly regions, road construction is a major driver of environmental damage. Excavation and hill cutting often trigger slope instability, landslides, and rockfalls. It also leads to deforestation, soil erosion, and the disruption of natural hydrological patterns.
- ii. **Safety Risks and Accidents:** Road transport is associated with a high frequency of accidents. In urban centers like Kathmandu, accidents are frequently caused by human factors (over-speeding, reckless or drunk driving) and poor road conditions (potholes, narrow lanes, and inadequate lighting).
- iii. **Vulnerability to Natural Calamities:** Roads are highly susceptible to being blocked or destroyed by landslides in the hills and flooding in the Terai plains, which can cut off connectivity for long periods.
- iv. **Traffic Congestion:** Rapid and unplanned urbanization has led to severe traffic congestion in cities, resulting in increased travel time, noise, and air pollution.
- v. **Limited Lifespan and Distress:** Pavements are subject to various types of distress, such as rutting, alligator cracking, and mud pumping, which require constant monitoring and periodic rehabilitation to remain serviceable.

The selection of a highway alignment is governed by topographic, geological, economic, and social factors. A well-chosen alignment ensures safety, minimizes construction and maintenance costs, and reduces environmental impact.

The factors that affect the selection of highway alignment are listed below:

I. Obligatory Points:

Obligatory points are those control points governing the highway alignment that tells if the alignment must pass through or should be avoided because of engineering, economic, social, or environmental considerations.

- **Points to Pass Through:** The alignment may be deviated to connect intermediate towns, reach suitable bridge sites (where the river is straight and banks are stable), or cross mountain passes to attain necessary height.
- **Points to Avoid:** To minimize costs and conflicts, the alignment should avoid religious places, costly permanent structures, lakes/ponds, and marshy or water-logged lands.

II. Topography:

- In mountainous regions, roads should follow natural ground contours as closely as possible to minimize massive excavation and filling.
- Design standards and speeds are determined by the cross-slope, categorized as Plain (0–10%), Rolling (10–25%), Mountainous (25–60%), or Steep (>60%).
- A primary goal is to achieve a balancing depth between cutting and filling to lower construction costs and reduce the need for external borrow pits.

III. Geological stability:

- Alignments must strictly avoid active or old landslide zones, rockfall-prone areas, and weak geological formations such as faults or shear zones.
- Excessive cutting through hard solid rock should be avoided to control project time and expenses.

IV. Geometric design:

- **Design Speed:** This is the most critical factor, as it determines curve radii, transition lengths, and required sight distances.
- **Gradients:** Alignments should maintain gentle gradients (desirably 5–7%). Where steep height must be gained in a short horizontal distance, hairpin bends or loops are introduced.
- **Sight Distance:** The alignment must ensure adequate Stopping Sight Distance (SSD) at all points, with Overtaking Sight Distance (OSD) provided where terrain allows.

2. **The radius of a horizontal highway curve is 450 m, super-elevation provided is 1 in 15 and the width of pavements curve is 7.5. If the rate of change of centrifugal acceleration is not to exceed 0.45 m/sec² and the rate of introduction of super elevation (about the inner edge of pavement) is not to exceed 1:150, design the length of horizontal transition curve for a design speed of 100 kmph.**

Answer:

Given,

Radius of horizontal curve, $R=450\text{m}$

Super-elevation, $e= 1/15$

Pavement width, $B= 7.5\text{m}$

Design speed, $V= 100\text{ km/hr}$

Maximum rate of change of centrifugal acceleration, $C= 0.45\text{m/s}^3$

Maximum rate of introduction of super-elevation = 1/150

We need to calculate the transition length based on both criteria and adopt the larger value.

Length based on rate of change of centrifugal acceleration:

$$\text{Design speed } \in \text{ m/s: } V = \frac{100 \times 1000}{3600} = 27.78 \text{ m/s}$$

$$\text{The length of transition curve is: } L = \frac{V^3}{CR} = \frac{21433.6}{202.5} = 105.85 \text{ m}$$

Length based on rate of introduction of super-elevation:

Since the super-elevation is introduced about the inner edge of the pavement, the outer edge has to be raised through:

$$h = eB$$

where

$$e = 1/15, B = 7.5 \text{ m}$$

Therefore,

$$h = \frac{1}{15} \times 7.5 = 0.5 \text{ m}$$

The permissible rate of introduction is 1/150.

Hence,

$$\begin{aligned} L_2 &= 150 \times h \\ L_2 &= 150 \times 0.5 \\ L_2 &= 75 \text{ m} \end{aligned}$$

Adopt the larger value:

$$\begin{aligned} L &= \max(L_1, L_2) \\ L &= \max(105.85, 75) \end{aligned}$$

Therefore,

$$\therefore L \approx 106 \text{ m}$$

3. **Explain in detail the special points that must be considered in the alignment of hill road of Nepal.**

Answer:

Hilly region of Nepal covers the mountainous & steep terrain having a cross slope of 25 to 60% and more. The road passing through the hilly terrain with a cross slope of 25% or more is generally termed as Hill Road. A hill road usually consists of either a river route or a ridge route. Road alignment in Nepal's hilly terrain is governed by a combination of geological, topographic, drainage, economic, and geometric design factors. The steep, unstable slopes and complex geology of the Himalayas and Middle Hills make the design and construction of hill roads more critical.



i. Geological Stability:

- Road alignment should pass through stable hill slopes and should avoid landslide and rockfall prone zones.
- Construction activities might trigger landslide and rockfalls if the road alignment is to be passed through the area having unstable or weak hill slopes.
- Excessive cutting works through hard solid rocks during construction must be avoided.

ii. Cross drainage structures:

- Hilly region consists of large number of fast flowing, steep rivers that requires numerous cross-drainage works.
- The road alignment should be selected such that the roadway consists of minimum number of cross drainage structures.

iii. Geometric Design:

- Geometric design of hill roads must be done according to the design guidelines and standards for hill region.
- The alignment should have minimum number of sharp curves and hairpin bends, gentler gradient.

iv. Availability of construction materials:

- The road alignment should be selected in such a way that the construction materials are available near the construction site.
- It reduces the transportation cost of materials and makes the construction more economical.

v. Environment and weather:

- Rainfall (or Snowfall) $\propto$ Altitude
- As the altitude decreases, the number of cross-drainage works required increases.
- At higher altitudes, the road pavement may witness snowfall during winter. This is why the alignment of hill roads should preferably be provided at an altitude between 900 m slopes exposed to sun.

vi. Length:

- The cost of construction of a hill road per kilometer length is comparatively very high.

- Hence, it should be ensured that the length of the road connecting two stations should be minimum possible.

4. **a. What are the general requirements of traffic control devices?**

b. What are traffic islands?

c. Briefly describe various types of traffic islands.

Answer:

a. Various devices that are used to control, regulate and guide traffic are known as traffic control devices. These devices provide safe and efficient functioning of intersections by separating conflicting vehicle streams in time.

There are different types of traffic control devices needed at the intersection. They are:

- Traffic sign
- Traffic signals
- Traffic island
- Road markings

The general requirements of traffic control devices are:

- A traffic control device should be installed only when it is necessary and justified by traffic conditions.
- The message should be simple, clear and easily understood without causing confusion.
- The design, color, shape, size and application of devices should be uniform so that road users can understand them quickly.
- The size of signs, markings and signals should be appropriate for the speed and traffic conditions of the road.
- Signs and markings should use clear symbols, lettering and colors that can be read and understood easily.
- Traffic Control devices should be regularly maintained so that they remain clean, visible, legible and effective.

b. Traffic island is the raised area provided at the level road to avoid or minimize the areas of major and minor conflicts. Island establishes physical channels through which vehicular traffic may be guided to travel and avoid conflict with other vehicular traffic. The functions of traffic islands are listed below:

- Traffic island separate and channelize traffic into desired directions.
- They provide a location for traffic signs, signals and road markings.
- They also help in reducing the effective width of the carriageway, thereby helping to control vehicle speeds.
- Islands improve road safety and reduce accidents.

c. Traffic island is the raised area provided at the level road to avoid or minimize the areas of major and minor conflicts. Traffic islands are of four types:

i. Divisional islands:

This type of traffic island separated traffic moving in opposite directions or separates different traffic streams. It is also called Median Island. The possibilities of head on collision and other accidents are reduced. To eliminate the head light glare during night driving, the width of this island should be quite large.

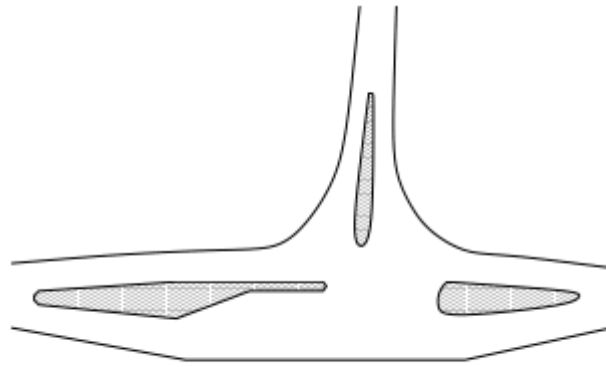
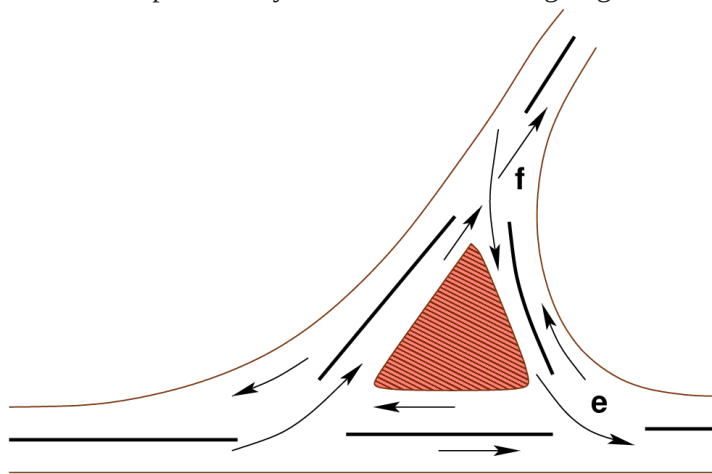


Figure 31:19: Divisional Islands

ii. Channelizing islands:

It guides vehicles into their proper paths at intersections and reduces conflicts between traffic movements. These islands are normally triangular in shape. They should be so located with respect with respect to approach that their visibility is clear and the intended path is automatically selected. They are very useful as a traffic control device particularly at intersections having large areas.

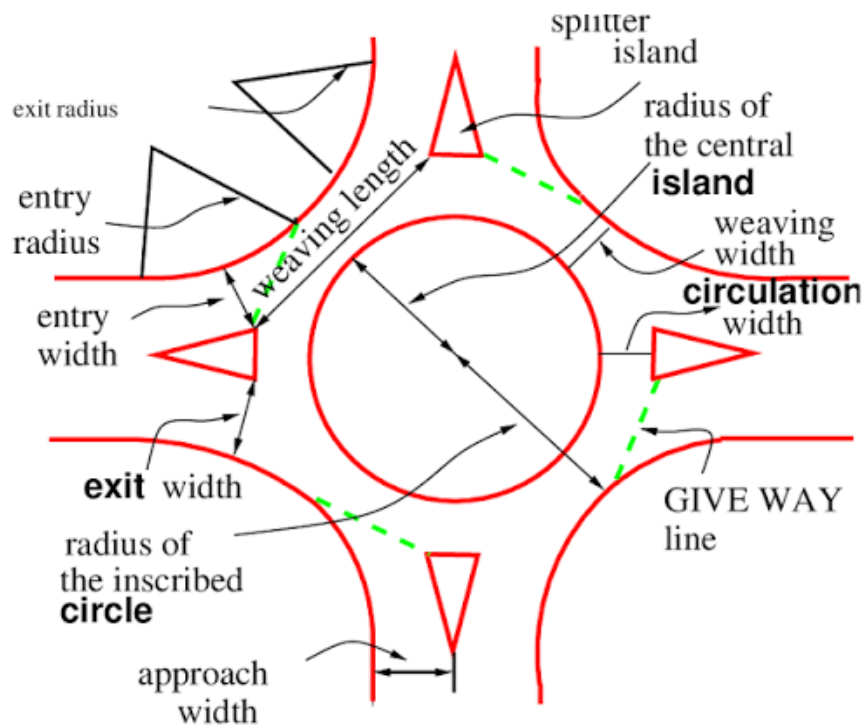


iii. Pedestrian loading islands:

It provides a safe waiting area for pedestrians crossing a wide or busy road, allowing them to cross in stages. They are provided at the regular bus stops and similar places for the safety of passenger. These are raised portion of road used by passenger either for riding on buses or loading of goods.

iv. Rotary island:

It is the central island around which vehicles circulate in a roundabout. Traffic rotary is an enlarged highway intersection where all the approaching vehicles from all the converging roads are forced to move around a large centrally situated island before they weave out into the desired radiating roads.



v. Refuse island:

They are located on the crosswalks to provide the refuse for pedestrians. On wide intersection approaches where the traffic flow is heavy refuse islands are commonly used.



5. **a. What is EAL or ESA? Describe its significance in design of road pavements.**
b. What are the various methods of design of road pavements?

Answer:

a. EAL (Equivalent Axle Load): It is the equivalent load effect of a particular vehicle axle expressed in terms of a standard axle load.

ESA (Equivalent Standard Axle): It is the number of standard axle-load repetitions that would produce approximately the same damaging effect on a pavement as the actual traffic loading.

EAL (Equivalent Axle Load) and ESA (Equivalent Standard Axle) are related but slightly different terms in pavement design. In pavement design, different axle loads are converted into equivalent standard axle repetitions so that the total traffic loading can be represented by a single value.

$$ESA = \sum n_i \left(\frac{W_i}{W_s} \right)^4$$

where:

n_i = number of repetitions of axle load W_i
 W_i = actual axle load
 W_s = standard axle load

The exponent represents the approximate fourth-power relationship between axle load and pavement damage.

Its significance in pavement design is listed below:

- i. ESA converts different axle configurations and loads of different vehicles into a common standard axle unit.
- ii. Traffic expressed in terms of cumulative standard axles helps in the design of flexible and semi-rigid pavements.
- iii. The cumulative ESA is used to estimate the total traffic loading that the pavement must withstand.
- iv. Higher cumulative ESA generally requires a stronger and thicker pavement structure.
- v. Different traffic compositions can be compared on the basis of their equivalent standard axle repetitions.

b. Road pavement can be defined as a stable layer constructed over the natural soil subgrade for supporting and distributing the wheel loads and providing an adequate surface for the movements of vehicles with certain speed safely, comfortably and economically.

The design methods for flexible pavements are listed below:

- i. Theoretical methods or mechanistic methods:
 - This method is based on the mechanics of pavement materials.
 - Wheel loads are used to calculate stresses, strains and deflections in different pavement layers.
 - Boussinesq's theory is used to determine stress in homogeneous mass and Burmister's theory in layered system.
 - The pavement is designed such that the critical stresses and strains remain within permissible limits.
 - It can account for different materials; axle loads and pavement configurations.
- ii. Semi-empirical methods or tri-axial methods:
 - This method combines theoretical analysis with empirical relationships.
 - The stresses and strains in pavement layers are estimated using theoretical principles.
 - Empirical relationships are then used to relate these stresses/strains to pavement performance.
 - It provides better reliability than purely empirical methods.
- iii. Empirical method:
 - This method is based on observations, experiments and past performance of existing pavements.
 - Pavement thickness is related to parameters such as traffic and subgrade strength.
 - It is simple and economical.
 - Its applicability is limited to conditions similar to those under which the empirical relationships were developed.
 - For example: GI method, CBR method etc.

6. Why road maintenance is necessary? Describe different type of road maintenance. Explain the maintenance of the bituminous pavement.

Answer:

Highway Road maintenance is the task of preserving, repairing and restoring a system of roadways to keep the serviceable conditions of highway as normal as possible and best as practicable. It is importance as maintenance helps in providing better service facilities longer life and better appearance. It involves variety of planning, programming and scheduling to actual implementation in the field and monitoring.

The road maintenance is necessary because of the following points:

- i. Highway maintenance ensures road safety after the repair of potholes, damaged shoulders, and reduced risk of road accidents.
- ii. It also extends the life of pavement by timely maintenance, preventing minor defects and cracks on it.
- iii. It also ensures smooth and continuous flow of traffic, reduces congestion and delay which maintains efficiency of the pavement.
- iv. Routine and preventive maintenance also reduces costly reconstruction and rehabilitation.
- v. Highway maintenance ensures effective drainage system as it comprises of cleaning and maintaining drains and culverts which prevents waterlogging.

Highway Maintenance activities can be classified into three categories as defined by DoR in DoR strategy (2016):

i. Routine maintenance activities:

- Those activities that are carried out as frequently as required during each year on all elements of the highway to improve its serviceability in all weather conditions are routine maintenance activities.
- Includes:
 - a. Grass and bush cutting, grading, reshaping of the unpaved surfaces.
 - b. Cleaning of carriageway, drains, slopes, culverts.
 - c. Replacement of structures that has been damaged.



ii. Recurrent maintenance activities:

- Those maintenance activities of localized nature carried out more or less at regular intervals of six months to 2 years are recurrent maintenance activities.
- Includes:
 - a. Sealing cracks, local surface treatment
 - b. Repair of depression and ruts
 - c. local constructions



iii. Periodic maintenance activities:

- Those activities done at the interval of several years to improve the overall functionality of road are periodic maintenance activities.
- Includes:
 - a. Renewal of wearing course of the pavement surface.
 - b. Restoration of road markings
 - c. Repainting of metal bridges etc.



iv. Special maintenance activities:

- Those activities of specialized and emergency nature done to solve special problems which may cause pavement failure if not identified on time.
- Includes:
 - a. construction of temporary diversion to allow the traffic to pass through obstruction.
 - b. Removal of debris and other obstacles.
 - c. Strengthening of pavement structures.

Road maintenance can also be grouped as:

i. Preventive maintenance:

It included actions that are taken to prevent premature deterioration and to retard the progression of deficiencies to reduce the rate of deterioration and increases useful life of pavement.

ii. Corrective maintenance:

It includes the action that includes actions taken to correct the deficiencies which are hazardous which seriously affect a pavement's operation to keep the highway within a tolerable level of serviceability.

Bituminous pavement maintenance involves regular inspection, preventive measures and repair works carried out to preserve the pavement, maintain riding quality, ensure safety and extend its service life. The maintenance of the bituminous pavements is described below:

i. Routine maintenance:

Routine maintenance is carried out regularly to prevent minor defects from becoming serious.

- Cleaning of pavement and shoulders.
- Cleaning and repairing of side drains and culverts.
- Removal of vegetation from pavement edges.
- Filling of minor cracks and potholes.
- Maintenance of road furniture and markings.

ii. Preventive maintenance: Preventive maintenance is performed before major deterioration occurs.

- Crack sealing: Seal cracks to prevent water penetration.
- Surface dressing: Application of bituminous binder followed by aggregate to restore the surface and improve waterproofing.
- Fog seal: Application of a thin bituminous coating to seal minor cracks and rejuvenate a dry surface.
- Slurry seal/micro-surfacing: Application of a thin mixture of bitumen, fine aggregate and additives to improve surface condition.

iii. Corrective maintenance:

Corrective maintenance is undertaken when pavement defects have already developed.

a. Pothole repair:

The damaged material is removed, the area is cleaned and dried, and suitable bituminous material is placed and compacted.

b. Patch repair:

Localized failed areas are cut out and replaced with new bituminous material.

c. Crack repair:

Depending on the severity, cracks are sealed or the affected pavement layer is removed and reconstructed.

d. Rut repair:

Ruts are corrected by leveling, patching or resurfacing the affected portion.

iv. Periodic maintenance:

When the pavement surface has deteriorated significantly but the underlying pavement structure is still adequate, a new bituminous layer is provided.

- Bituminous overlay is commonly used.
- It restores riding quality and waterproofing.
- It increases the service life of the pavement.

v. Rehabilitation and reconstruction:

If the pavement has severe structural failure, simple surface maintenance is insufficient.

- Failed pavement layers are removed and reconstructed.
- Base/sub-base may be strengthened or replaced.
- Drainage problems are corrected.
- A new bituminous surface is then provided.

7. **a. What are the different types of landslides? Describe them briefly.**

b. How can slope stability be improved by plants?

Answer:

a. Landslide is the movement of a mass of soil, rock or debris down the slope under gravity. It is caused when gravitational pull exceeds the strength of soil. In the context of Nepal, rugged topography, unstable hills with large unstable slopes are commonly found in mountain and hilly regions. These regions report frequent landslides especially in mid-hill regions, Chure and Mahabharat range. More than 80% of the territory is characterized by steep hills and mountains prone to landslides. Few examples of the occurrence of landslide are Jure landslide, Sindhupalchowk, Bhirkot village landslide, Dolakha, Mugling-Naraynghad highway landslide disasters and so on.

The different types of landslides are listed below:

i. Fall:

It occurs when rock or soil detaches from a steep slope or cliff and moves rapidly through the air by falling, bouncing or rolling. It is very common in steep rocky terrain. It is usually sudden and difficult to predict.

ii. Topple:

A topple occurs when a block of rock rotates or tilts forward about a point or axis below its centre of gravity. It is common where rocks have steeply inclined joints or fractures.

iii. Slides:

A slide occurs when a mass of soil or rock moves downward along a relatively distinct failure surface. There are two types of slides: rotational slide and translational slide.

iv. Flows:

In a flow, the displaced material behaves like a viscous fluid and moves continuously down the slope. Examples: mudflow, debris flow, earth flow etc.

v. Creep:

Creep is very slow, gradual downward movement of soil or rock under gravity. It may occur over years or decades. It is often indicated by tilted trees, poles, fences and cracked walls.

vi. Lateral spreading:

Lateral spreading occurs when a relatively coherent soil mass moves horizontally over a weak or liquefiable underlying layer. It is often associated with earthquake-induced soil liquefaction.

b. Slope stability can be significantly improved through bio-engineering. Stabilization of slope using vegetative systems with small civil engineering structures is known as bio-engineering. It combines biological measures (plants) with civil engineering works to improve slope stability in an environmentally friendly and cost-effective manner.

Example:

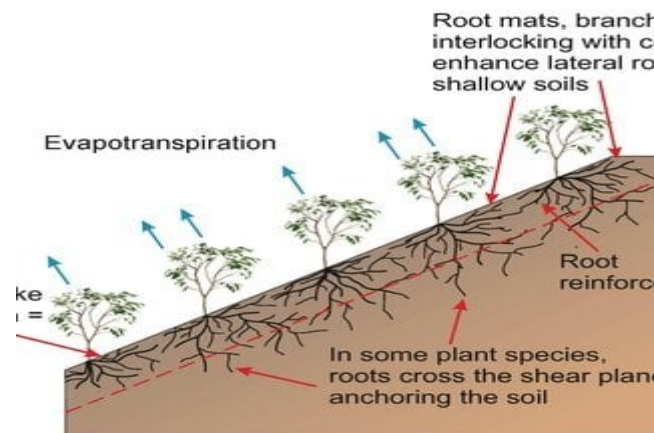
i. Toe wall with bamboo plantation:

ii. Jute net with grass seedling:

This approach is particularly effective in Nepal's geologically fragile hilly and mountainous regions. Plants improve slope stability by:

i. Root Reinforcement:

Plant roots act as natural anchorage, binding soil particles together to increase the soil's overall shear strength. Deep-rooted shrubs and trees are especially effective at reinforcing deeper soil layers.



ii. Lowering Pore Water Pressure:

Plants absorb soil moisture through their root systems, which directly lowers internal pore water pressure. High pore water pressure is a primary trigger for landslides as it reduces the effective stress and shear strength of the soil.

iii. Erosion Protection:

Vegetative cover reduces the direct impact of rainfall (rain splash) and slows the velocity of surface runoff. This helps prevent the formation of gullies and cross ruts on road embankments.

iv. Weathering Prevention:

A natural layer of vegetation protects exposed slopes from weathering and structural deterioration caused by harsh climatic conditions.

8. a. What is tunneling? Briefly describe the ventilation and lighting requirements for tunnels.

b. If you are given an opportunity to select an ideal bridge site, what do you consider? And why?

Answer:

a. The process of construction of an underground passage through soil or rock, for transportation routes, water conveyance, utilities etc. is known as tunneling. The option of tunnel is opted when underground passage is

more economical than excavation and cutting. Numerous tunnels are constructed in Nepal for hydropower projects, irrigation projects, water supply projects and very few for road transport. Examples: Bheri-Babai Diversion Multipurpose project with TBM, Nagdhunga tunnel with NATM, Melamchi water supply with Drill and Blast Method and so on.

The ventilation requirement for tunnels are as follows:

- i. Harmful gases such as carbon monoxide and other pollutants from the vehicles or construction equipment must be removed.
- ii. Adequate fresh air should be supplied throughout the length of tunnel.
- iii. Proper air circulation should maintain the temperature and humidity condition under control.
- iv. Air movement should be sufficient for effective ventilation without creating uncomfortable conditions.
- v. The system should be capable of controlling and removing smoke during fire or other emergency.

Adequate lighting is essential because natural daylight is limited or absent inside a tunnel. The lighting requirements for tunnel are as follows:

- i. The entrance area should have adequate lighting to help drivers adjust from daylight to tunnel conditions.
- ii. Uniform illumination should be maintained throughout the main tunnel section.
- iii. Adequate lighting should be provided near the exit to avoid sudden changes in visibility.
- iv. Emergency lights should be present in case of power failure.
- v. Sudden dark and bright patches must be avoided inside the tunnel.

b. If I am given an opportunity to select an ideal bridge site, the following factors should be considered:

- i. The site should be located on a straight and stable reach of river.
- ii. Relatively narrow section is selected as it reduces the length of the bridge and also the cost of construction required.
- iii. The river beds and banks must have strong and stable strata capable of supporting bridge foundations and resisting scour.
- iv. The banks should be sufficiently high and stable to reduce the risk of flooding and erosion near the approaches.
- v. The bridge site should be free from active faults, landslides, unstable slopes and other geological hazards.
- vi. It should be selected where there is sufficient availability of construction materials, labor, machinery, access roads and so on.
- vii. The bridge site should minimize the total cost of the bridge, approaches, foundations and protection works.

9. Define active and passive earth pressure in soil. Derive an expression for active and passive earth pressure by Rankine's method.

Answer:

The minimum lateral pressure exerted by the soil on the wall, when it moves away from the backfill causing the soil behind it to expand reaching the active state is called active earth pressure.

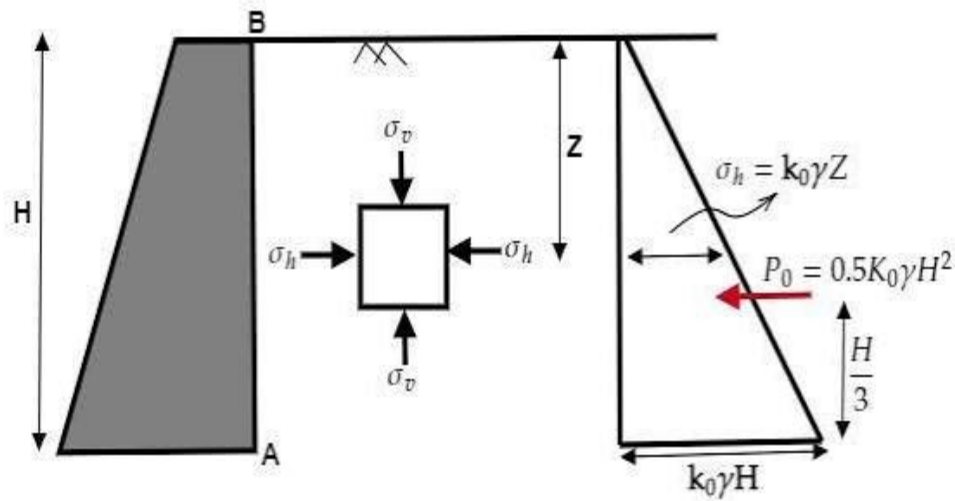
The maximum lateral resistance offered by the soil when a retaining wall moves towards the backfill soil, compressing the soil and reaching the passive state is called passive earth pressure.

The expression for active earth pressure on the vertical back face of earthwork supporting granular soil having horizontal surface is given by Rankine's theory.

The assumptions for Rankine's theory:

- i. The back of the retaining wall is vertical and smooth (no wall friction).
- ii. The backfill is dry, homogeneous, cohesionless (granular) soil.
- iii. The ground surface is horizontal.
- iv. The soil is in the plastic equilibrium state.

- v. Failure occurs along a plane surface.



Let us consider a soil element at a depth h below the ground surface.

Vertical stress due to overburden:

$$\sigma_v = \gamma h$$

where:

$$\gamma = \text{unit weight of soil (kN/m}^3\text{)}$$

$$h = \text{depth below ground level (m)}$$

The horizontal stress under active condition is

$$\sigma_h = K_a \sigma_v$$

where K_a is the coefficient of active earth pressure.

According to Rankine's failure criterion,

$$K_a = \frac{1 - \sin \phi}{1 + \sin \phi}$$

or

$$K_a = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$$

where

ϕ = angle of internal friction of soil.

Substituting

$$\sigma_h = K_a \gamma h$$

For a retaining wall of height H ,

Pressure at the top:

$$p=0$$

Pressure at the bottom:

$$p = K_a \gamma H$$

The total active earth pressure equals the area of the pressure triangle.

$$P_a = \frac{1}{2} \times \text{Base} \times \text{Height}$$

$$P_a = \frac{1}{2} (K_a \gamma H) (H)$$

Therefore,

$$P_a = \frac{1}{2} K_a \gamma H^2$$

This force acts horizontally on the wall.

Since the pressure distribution is triangular, the resultant acts at the centroid of the triangle.

Hence,

$$\text{Point of application} = \frac{H}{3} \text{ above the base}$$

At passive failure, Rankine's passive earth-pressure coefficient is:

$$K_p = \frac{1 + \sin \phi}{1 - \sin \phi}$$

∴,

$$K_a = \tan^2 \left(45^\circ + \frac{\phi}{2} \right)$$

The lateral passive pressure at depth z is:

$$\sigma_{hp} = K_p \gamma z$$

At the bottom of the wall:

$$\sigma_{hp, \max} = K_p \gamma H$$

The total passive earth pressure is therefore:

$$P_p = \frac{1}{2} \gamma H^2 \tan^2 \left(45^\circ + \frac{\phi}{2} \right)$$

Hence,

The resultant P_p also acts at $\frac{H}{3}$ above the base.

10. a. Describe briefly the design procedure of a mat foundation using conventional method of design.
b. How do you determine linear water way for a bridge to be constructed in an alluvial plain? What will happen if the linear water way is not sufficient?

Answer:

a. Raft foundation also known as mat foundation is the slab supporting large number of columns in two or more rows. It is used in compressible soil layer where isolated footings don't perform well and commercial buildings where isolated footing covers more than 50% of the plinth area.

The design procedure of a mat foundation using conventional method of design is explained below:

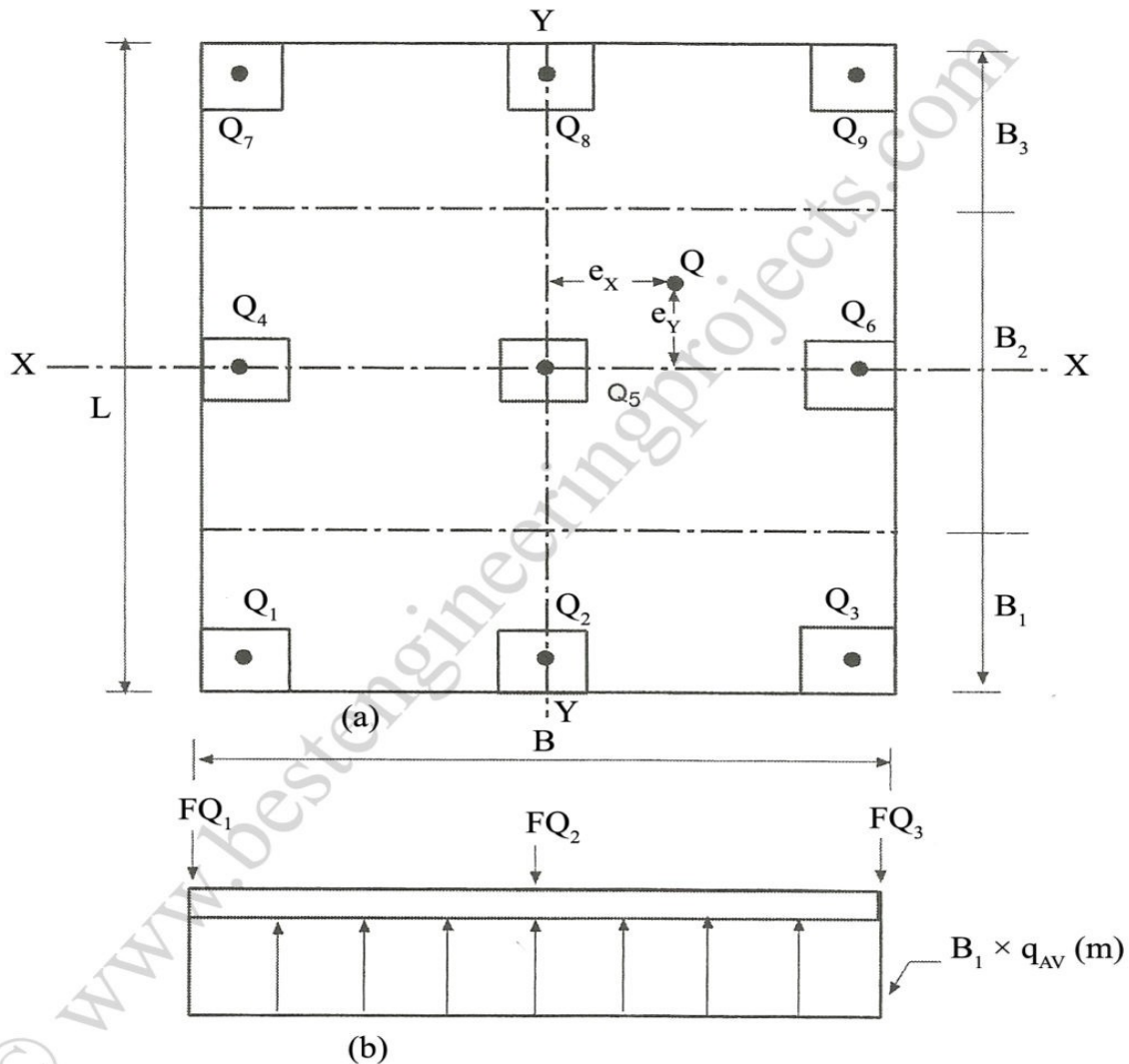


Fig 1 Design of Mat Foundation

- i. Collection of design data:
Required design data such as column loads, position of loads, column sizes, safe bearing capacity of the soil, depth of foundation etc. are collected.
- ii. Calculate C.G. of the load from column:
Taking moment about YY face

$$\bar{X} = \frac{\sum M}{Q}$$

Where, $Q = Q_1 + Q_2 + Q_3 + \dots$

Taking moment about XX face

$$\bar{Y} = \frac{\sum M}{Q}$$

iii. Calculation of eccentricity:

$$e_x = \bar{x} - \frac{B}{2}$$

$$e_y = \bar{y} - \frac{L}{2}$$

iv. Stress developed below the mat under any point:

$$q = \frac{Q}{A} + \frac{Q \times e_x \times x}{I_{yy}} + \frac{Q \times e_y \times y}{I_{xx}}$$

$$I_{yy} = \frac{LB^3}{12}, I_{xx} = \frac{BL^3}{12}$$

v. Divide the mat into number of beams in both direction:

vi. Convert each beam to equivalent beam and calculate the stress developed at each corner points of the mat.

For left corner M,

$$\bar{x} = -\frac{B}{2}, \bar{y} = +\frac{L}{2}$$

$$q_m = \frac{Q}{A} + \frac{Q \times e_x \times x}{I_{yy}} + \frac{Q \times e_y \times y}{I_{xx}}$$

For left corner R,

$$\bar{x} = +\frac{B}{2}, \bar{y} = +\frac{L}{2}$$

$$q_r = \frac{Q}{A} + \frac{Q \times e_x \times x}{I_{yy}} + \frac{Q \times e_y \times y}{I_{xx}}$$

$$\bar{q} = \frac{q_m + q_r}{2}$$

$$q_{avg} = \frac{1(\text{downward} + \text{upward})}{2}$$

$$= \frac{1[(Q_1 + Q_2 + Q_3) + \bar{q} \times B \times B']}{2}$$

$$= \frac{q_{avg}}{B \times B'}$$

vii. Calculate bending moments and shear forces.

viii. Draw SFD and BMD for each equivalent beam. Calculate maximum shear force and BM.

ix. Identify the largest maximum S.F. and B.M. among all the beams.

x. Design the thickness of slab and reinforcement using maximum S.F. and maximum B.M.

b. The total width of the opening of a bridge available for the passage of flood water, measured along the direction of the river flow is known as linear waterway. In an alluvial river, it should be sufficient to pass the design flood without causing excessive afflux, scour or changes in the river course.

We can determine the linear waterway for the bridge constructed in alluvial plain by following procedures:

i. The design flood discharge for the required return period is estimated using available hydrological data and suitable flood estimation methods.

- ii. A detailed river survey is carried out at the proposed bridge site including the river bed, banks and flood levels.
- iii. The existing river width and cross-sectional characteristics are studied. In an alluvial plain, the river adjusts its width and depth according to the sediment carrying capacity and discharge of river.
- iv. The required waterway may be estimated by using regime relations based on design discharge, flow velocity, hydraulic depth etc. and by using continuity equation. It is also given by Lacey's regime formula as, $P = 4.75 \sqrt{Q}$ and check the adequacy of the proposed waterway.

If the provided linear waterway is less than the required waterway, the bridge acts as an obstruction to flood flow. This can result in:

- Rise in upstream water level due to excessive afflux.
- Flooding of upstream areas and damage to nearby settlements and agricultural land.
- Scouring around piers and abutments, which may undermine foundations.
- Erosion of river banks and possible change in the river course.
- In severe cases, failure of bridge components or the entire bridge.